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Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Hannah Kettle MATHS TUTORIALS



Half Past Paper – Week 20

Higher Tier

Total Marks

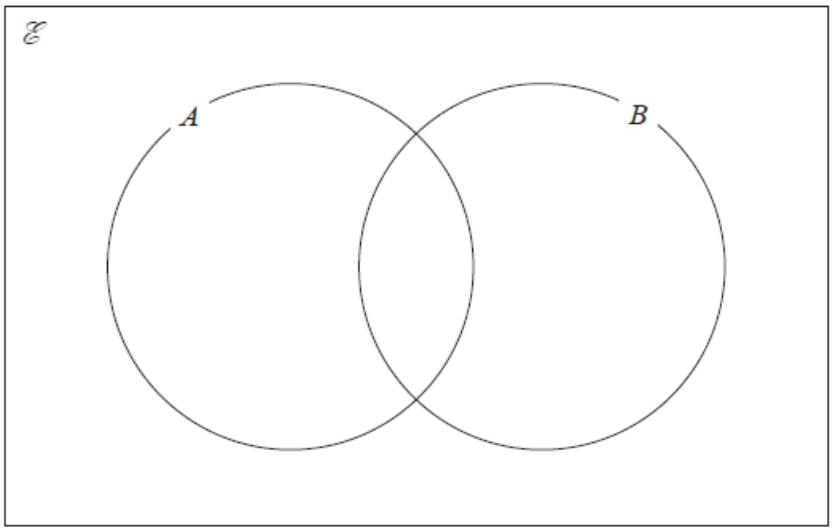
Q	Topic	Marks	Sparx Code
2	Venn diagrams, Using set notation	3	U476,U296
4	Percentage change without a calculator	4	U773
6	Finding the volume of prisms	3	U174
8	Factorising to solve quadratic equations of the form x^2 + bx + c = 0	3	U228
10a	Problem solving: Area and perimeter, Estimating calculations	3	U934,U225
10b	Estimating calculations	1	U225
12a	Drawing cumulative frequency graphs	2	U182
12b	Interpreting cumulative frequency graphs	1	U642
14	Problem solving: Area and perimeter of rectangles and compound shapes	3	U934
16	Tree diagrams for independent events	4	U558
18	Finding unknown sides in right-angled triangles, Sharing amounts in a given ratio	5	U283,U577
20	Equations of circles and tangents, Solving simultaneous equations involving quadratics	3	U567,U547
22	Finding the turning point of a quadratic graph by completing the square	4	U769
	Total	39	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

2 $\mathscr{E}$ = {even numbers less than 19}
A = {6, 12, 18}
B = {2, 6, 14, 18}
Complete the Venn diagram for this information.



(Total for Question 2 is 3 marks)

4 At the end of 2017

the value of Tamara's house was £220 000
the value of Rahim's house was £160 000

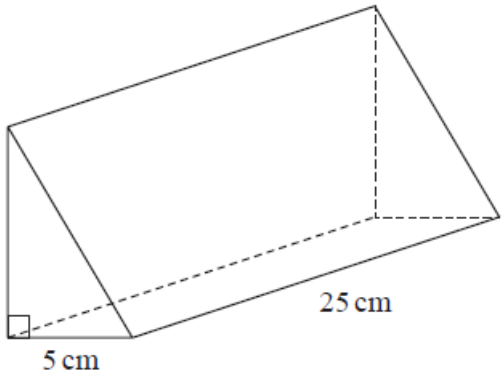
At the end of 2019

the value of Tamara's house had decreased by 20%
the value of Rahim's house had increased by 30%

At the end of 2019, whose house had the greater value?
You must show how you get your answer.

(Total for Question 4 is 4 marks)

6 The diagram shows a prism.



The cross section of the prism is a right-angled triangle.
The base of the triangle has length 5 cm

The prism has length 25 cm
The prism has volume 750 cm^3

Work out the height of the prism.

..... cm

(Total for Question 6 is 3 marks)

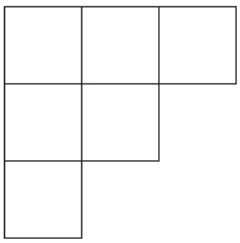
8 Solve $x^2 = 5x + 24$

.....

(Total for Question 8 is 3 marks)

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10 The diagram shows a shape made from 6 identical squares.



The total area of the shape is 5406 cm^2

(*a*) Find an estimate for the length of one side of each square.
Give your answer correct to the nearest whole number.

..... cm

(3)

(*b*) Is your answer to part (*a*) an underestimate or an overestimate?
You must give a reason for your answer.

.....

.....

.....

(1)

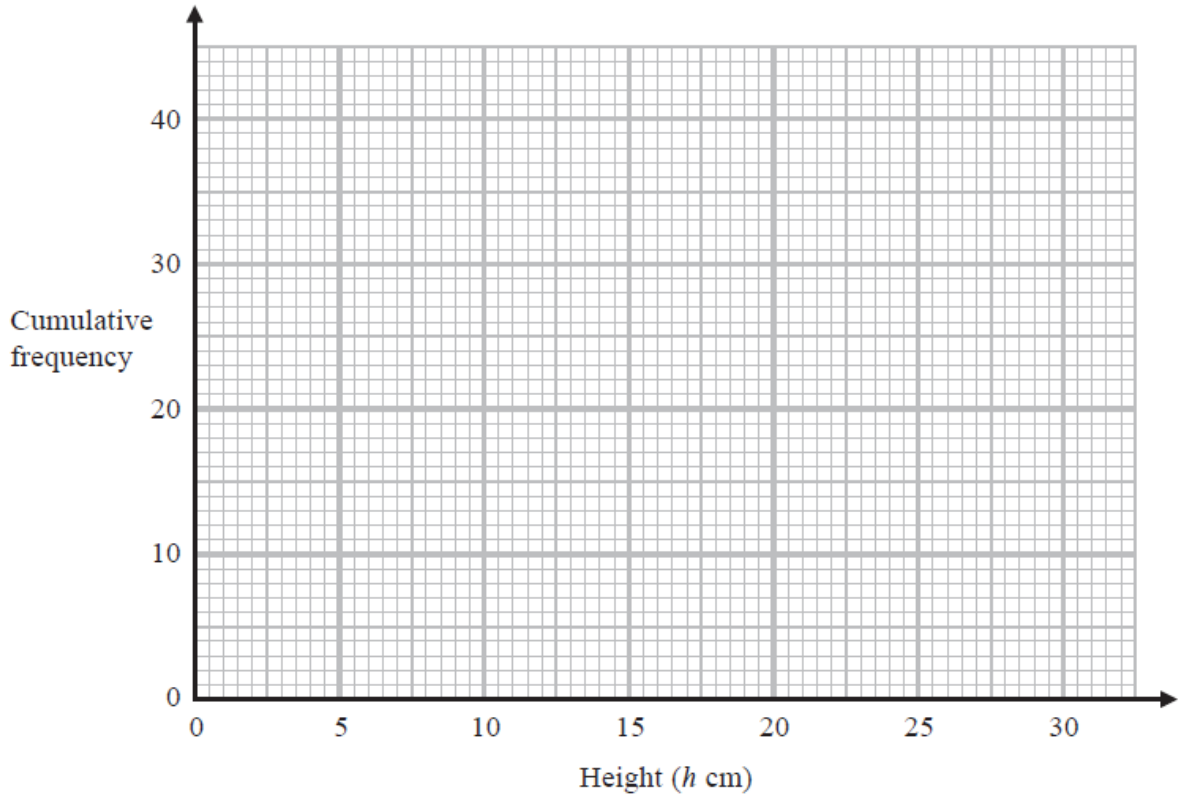
(Total for Question 10 is 4 marks)

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12 The cumulative frequency table gives information about the heights, in cm, of 40 plants.

Height (h cm)	Cumulative Frequency
$0 < h \leq 5$	4
$0 < h \leq 10$	11
$0 < h \leq 15$	24
$0 < h \leq 20$	34
$0 < h \leq 25$	38
$0 < h \leq 30$	40

(a) On the grid, draw a cumulative frequency graph for this information.



(2)

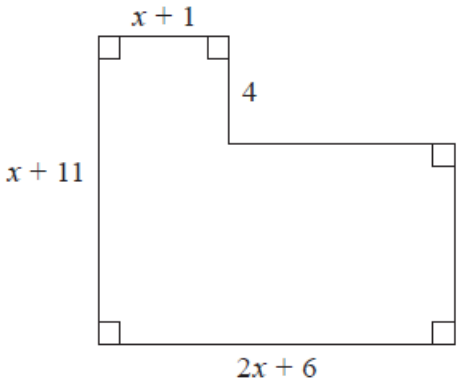
(b) Use the graph to find an estimate for the median height of the plants.

..... cm

(1)

(Total for Question 12 is 3 marks)

14 Here is a shape with all its measurements in centimetres.



The area of the shape is A cm^2

Show that $A = 2x^2 + 24x + 46$

(Total for Question 14 is 3 marks)

16 There are only 3 red counters and 5 yellow counters in a bag.

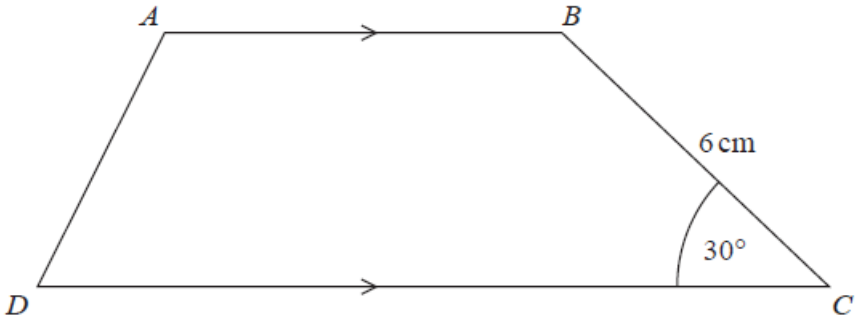
Jude takes at random 3 counters from the bag.

Work out the probability that he takes exactly one red counter.

.....

(Total for Question 16 is 4 marks)

18 Here is trapezium $ABCD$.



The area of the trapezium is 66 cm^2

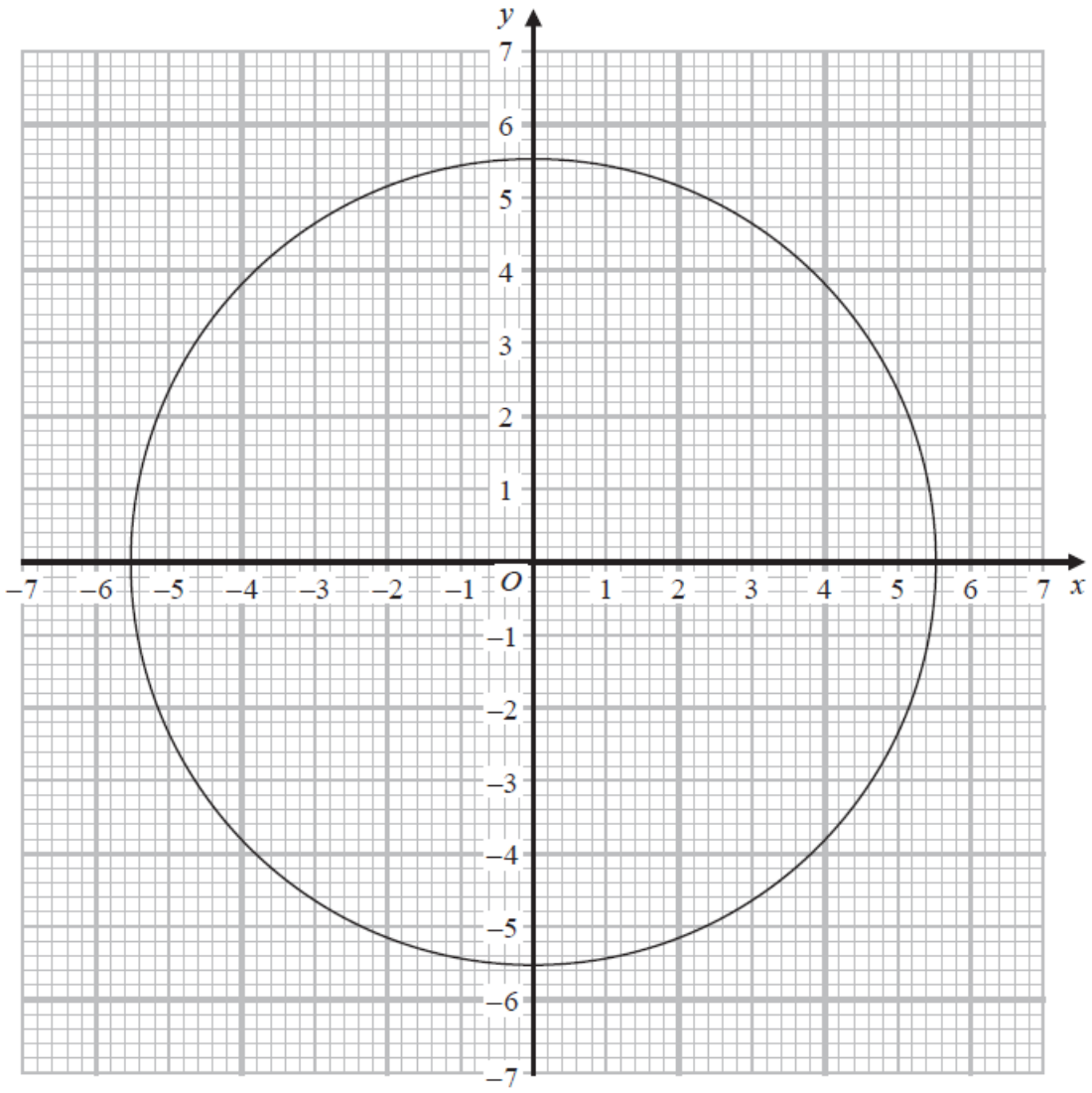
the length of AB : the length of CD = 2 : 3

Find the length of AB.

..... cm

(Total for Question 18 is 5 marks)

20 The diagram shows the graph of $x^2 + y^2 = 30.25$



Use the graph to find estimates for the solutions of the simultaneous equations

$$x^2 + y^2 = 30.25$$
$$y - 2x = 1$$

.....

(Total for Question 20 is 3 marks)

22 Find the coordinates of the turning point on the curve with equation $y = 9 + 18x - 3x^2$
You must show all your working.

(..... , ...)

(Total for Question 22 is 4 marks)

TOTAL FOR PAPER IS 39 MARKS